The New Exploration Tool: GenAI That Builds the Solutions You've Always Wanted

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Views expressed are my own

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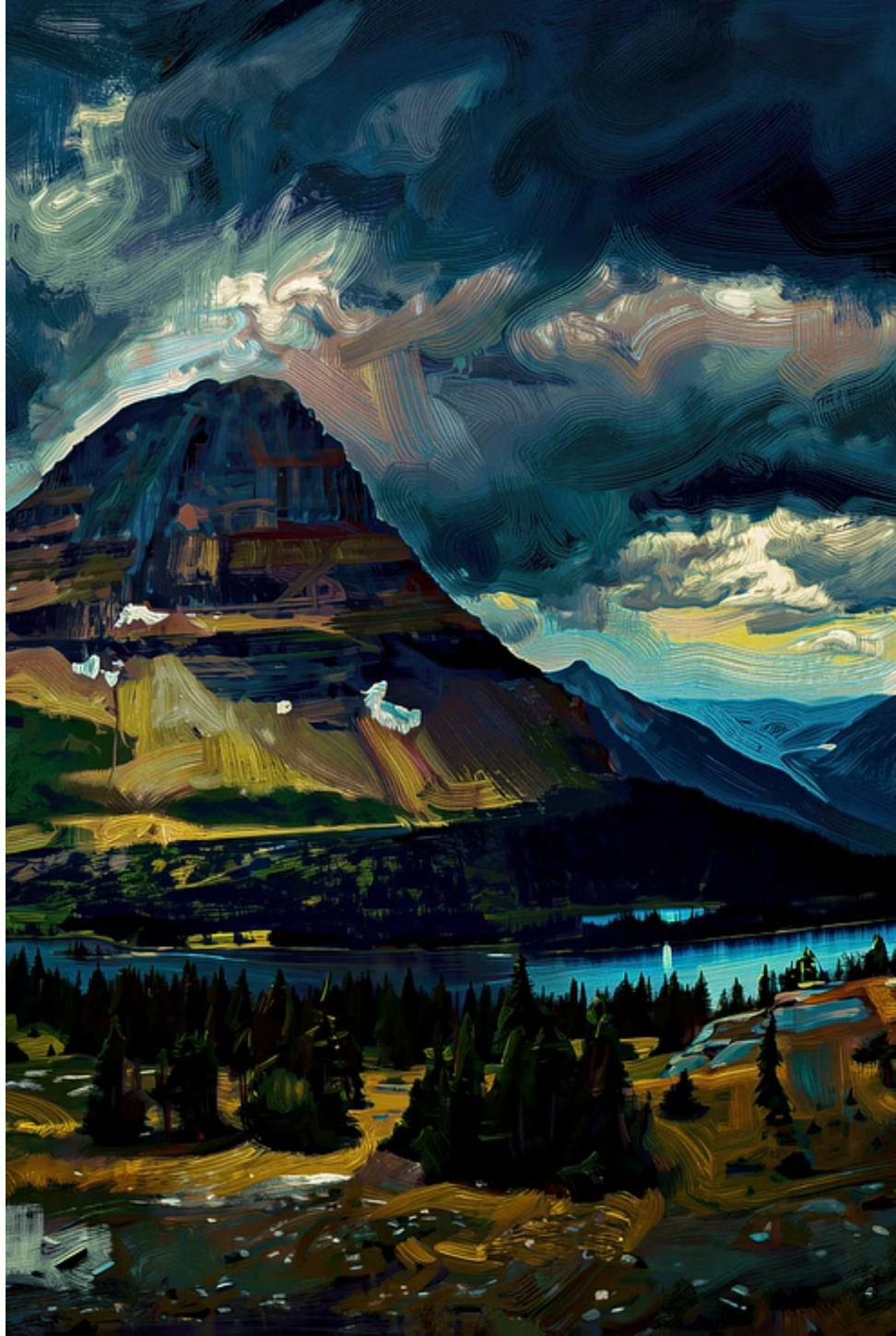
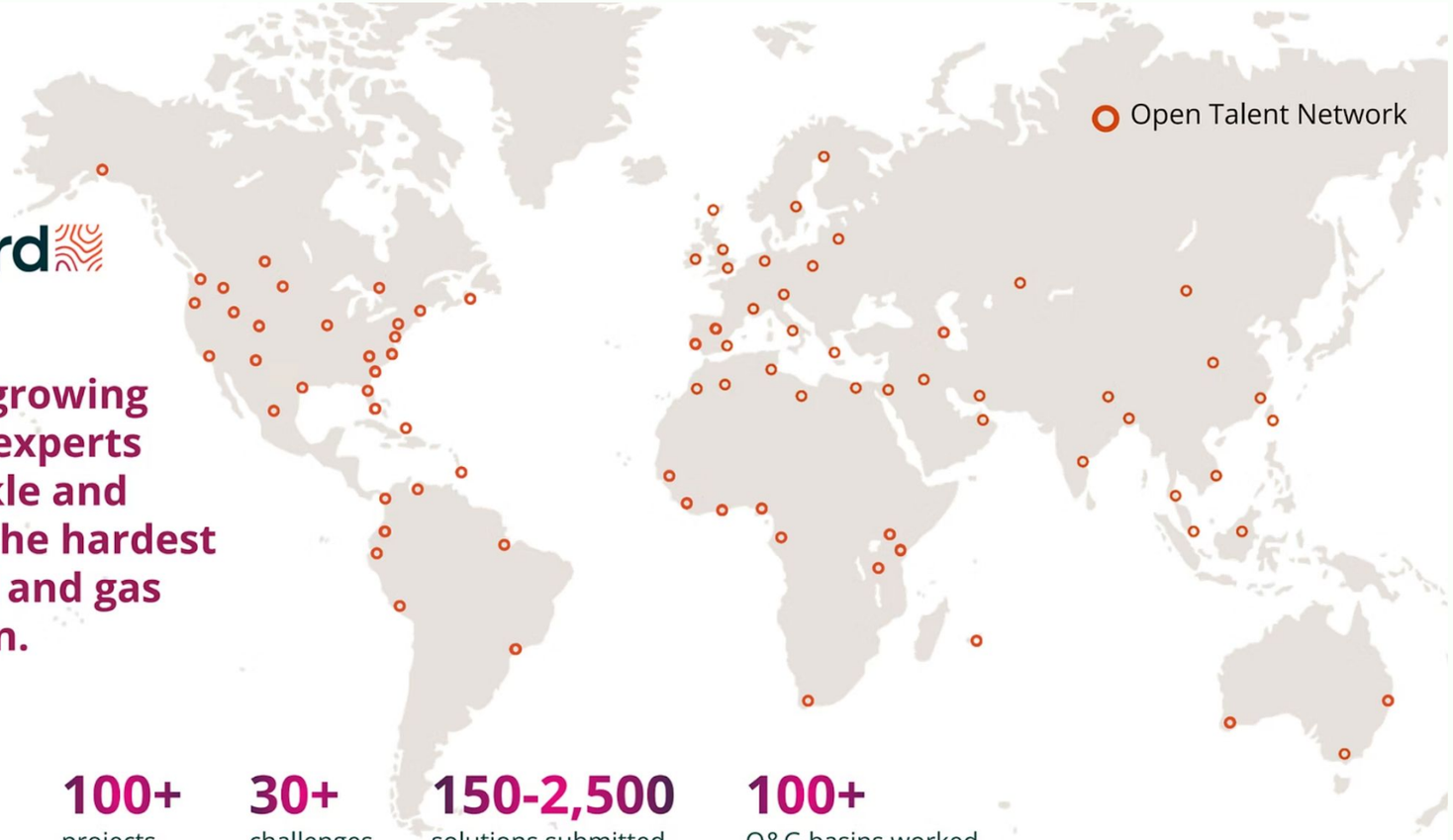


thinkonward

We utilize our growing community of experts (XiRs) who tackle and solve some of the hardest problems in oil and gas and exploration.



7,500+
curated geoscience & data science community members

100+
projects delivered

30+
challenges delivered

150-2,500
solutions submitted to every challenge

100+
O&G basins worked on by the community

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CHAPTER 1 CONTEXT

You've Been Here Before

Geoscientists have always adopted new tools. This is next.

~1980s
Personal computers & seismic processing

~1990s
Digital mapping & GIS

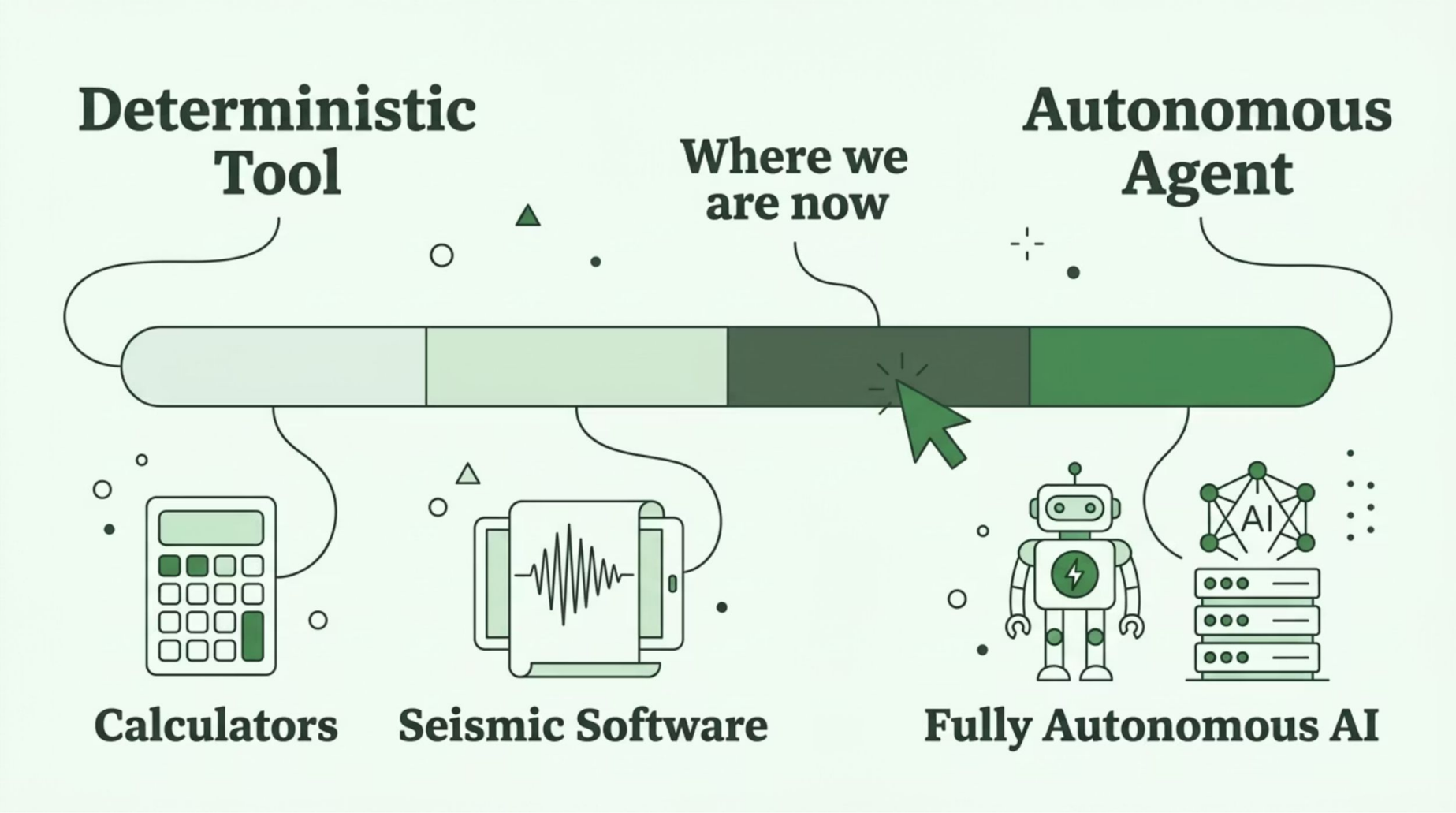
~2010s
FWI & advanced imaging

2023–Now
GenAI

Every tool shifted **how** we work; none replaced **why** we work

- Automated contouring didn't eliminate judgment – it required more of it
- GenAI is the next step in a lineage we already know
- We've been here before. We were fine.

This One Is Different, And That's Worth Saying Out Loud



Previous tools did exactly what you told them – GenAI **infers and fills gaps**

- It makes judgment calls – that's the power and the responsibility in one package
- Concerns about reliability and autonomy are valid – not reasons to disengage
- Eyes-open engagement beats fearful avoidance or uncritical adoption
- **The contouring map still has to make sense. Always.**

CORE THESIS

We Are Moving From a "How" World to a "What" World

- Most cognitive energy has gone to plumbing – formats, tools, pipelines

The scientific question – *what does this basin tell us* – gets whatever's left

- GenAI collapses the "how" so you can lead with the "what"

This is not a productivity improvement – it's a **reorientation** of where your mind lives

> **"GenAI doesn't replace your thinking. It gives you room to think."**

You've Spent Your Career Building Ideas Out of Incomplete Data, Through Layers of Abstraction. You Can Build Anything.

> Geoscience is a "hard to verify task", a task that will take the frontier for GenAI to solve for the next 10 years

In the "What" World, You Are the Bottleneck

> **"The models are ready. The question is whether you can describe what you want clearly enough to use them."**

- At current capability levels, compute and code are becoming less of a constraint
- Your ability to articulate ideas precisely and creatively limits output quality
- Geoscientists have spent careers communicating complex subsurface concepts – this is that skill
- Unclear prompt → mediocre output. Sharp problem definition → powerful result.
- **The most important thing you can practice right now is describing what you want**

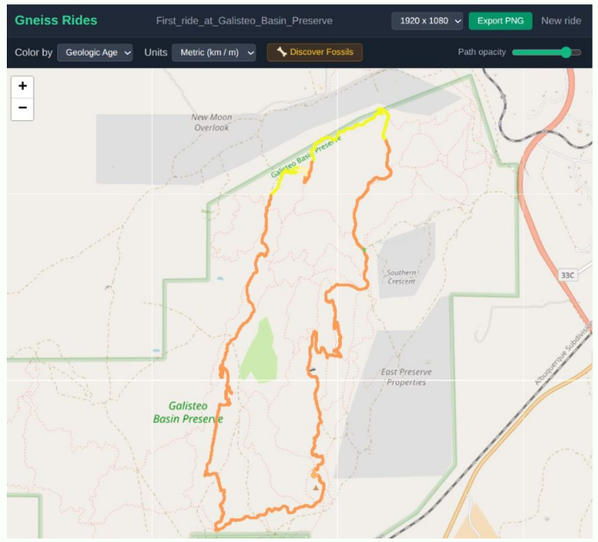
FUN APP

Gneiss Rides — App Walk-through

"What geologic age is the strata I'm about to crash my bike into?"

Map View

- Upload a GPX file, visualize your route
- Color path by Geologic Age or Formation
- Toggle "Discover Fossils" mode
- Export as PNG to brag on Strava
- Thank you Macrostrat



Geologic Cross-Section

- Stratigraphic elevation profile auto-generated along your route
- Formations color-coded by age
- Export as PNG

🌐 Try it live: https://nsuurmey.github.io/gneiss-rides/

EXPERIMENT

Block Party — Data Science Experiment

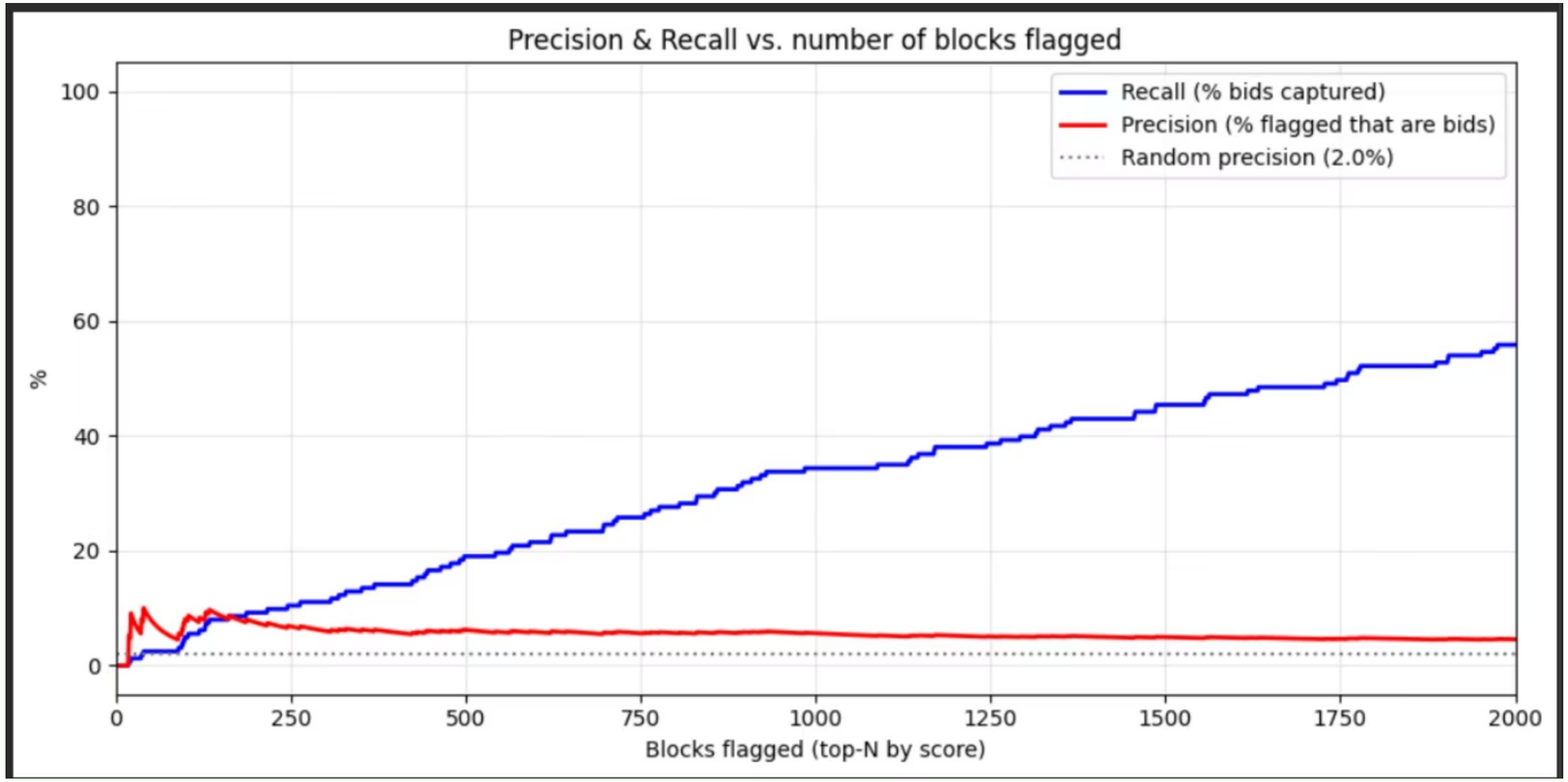
- **"What if there is a signal in the bid placement, block relinquishment, and well placement that could predict future operator bids in the Lease Sale?"**

- Note: Gemini agentic browsing is effective and scary.

🌐 Try it yourself: https://github.com/nsuurmey/block-party

TECHNICAL OUTPUT

Block Party — What the Model Actually Found



The Gray Line — Random Baseline

No model – pure guesswork. If you evaluated every block equally, roughly 2 out of 100 would turn out to receive bids. That's the haystack you're searching through.

The Red Line — Feature Signal

By ranking blocks on just two features – who leases next door and where rigs have been active – the top of the list is 3-4x more concentrated with eventual bids than a random draw.

The Blue Line — Operational Value

Flagging just the top ~1,000 blocks (roughly 12% of the sale universe) captures about a third of all blocks that actually received bids. You're cutting 85% of the noise without doing any subsurface work.

The Gap Between Lines

Bid rounds are sparse – hundreds of companies look at thousands of blocks and only bid on a few. No surface-level screen will be a bullseye, but it can tell you where to point the telescope.

Everything here is public data. The value isn't secret information – it's systematic prioritization of what everyone can already see but nobody is aggregating.

How to Think About Working with GenAI

The Phil Story

- Junior geoscientist, East Canada, needed a pore pressure model
- Asked a colleague → directed to Phil, the expert
- Executed iteratively, emailed Phil when stuck
- Phil QC'd – geoscientist caught the Jurassic anomaly from well report knowledge

The GenAI Version

- Needed an app, didn't know where to start
- Asked Claude → directed to write a PRD, then wrote it together
- Iterated with Claude, described what broke, asked what to do
- Validated against known basin, caught systematic GPS error from domain knowledge

> **The workflow was identical. The responsibility never left the room.**

GenAI Is Phil. You Still Have To Be You.

You

- Described the problem
- Iterated on the approach
- Caught what was wrong
- Knew what you were trying to do

GenAI (Phil)

- Scoped the project
- Guided implementation
- Implemented on request
- Asked Claude how to ask Claude – the "what" world in action

> **"The 'what' was always yours. Now you actually have time to think about it."**

Think of Your LLM as a Genie

Great power, real limits.

Answers exactly what you asked. Not what you meant.

- Genies are powerful but constrained – tokens are finite, context windows have limits (reduced performance after 50% saturation)
- You can't wish for infinite wishes – know your compute budget and work within it
- A genie grants the wish as stated, not the wish as intended
- Vague prompt → technically correct, completely wrong answer
- Specificity is the difference between magic and a monkey's paw

> **"The Genie will always answer. Your job is to ask the right question."**

The Anatomy of a Good Prompt

Layer 1 — ROLE

"You are a senior petroleum geoscientist evaluating well log data in the Permian Basin."

Layer 2 — CONTEXT & RESEARCH

"I have LWD logs from a horizontal lateral with inconsistent porosity readings in the Wolfcamp A."

Layer 3 — TASK

"Identify the three most likely causes and rank them by probability."

Layer 4 — CONSTRAINTS

"Plain language. 200 words max. No speculation beyond the data provided."

Layer 5 — CLARIFICATION TRIGGER

"Before you answer, ask me any clarifying questions that would improve your response."

> **"A vague prompt is a vague wish. The Genie answers exactly what you asked."**

The Build Workflow

Phase	Model	Learnings
PHASE 1 — SET UP YOUR PHIL | Assign a persona before you build. Match it to the task.	🔴 Premium model	A poorly defined collaborator gives you a poorly defined build.
PHASE 2 — DESCRIBE | Talk it out. Scope it ruthlessly. Generate a PRD.	🔴 Premium model	Most time spent here iterating.
PHASE 3 — ARCHITECT | Design the structure before anyone writes a line of code.	🔴 Premium model	Architecture decisions are expensive to undo. Spend the tokens now. Block Party v1 failed here, PRD too ambitious.
PHASE 4 — BUILD | One feature. One MVP. Never the whole thing at once.	🟢 Lightweight to medium model	One LLM session, One feature at a time.
PHASE 5 — FIX AND REPEAT | Things will break. That's the process, not a failure.	🟢 Lightweight model	If you've been in circles for 3+ cycles, back up and re-architect.
PHASE 6 — QC | No model. Just you. Non-negotiable.	🔵 Human only	The model will always give you an answer. Only you know if it's the right one.

> Use heavy models to think. Light models to build. A voice recorder to brain dump.

Where To Start: Tools and Techniques That Actually Work

Entry: NotebookLM

Load papers, reports, audio, video and ask questions, synthesize. **Go get this today.**

Intermediate: Frontier Lab Access

Set a spending cap, use for reasoning, drafting, MVPs

Advanced: AI-Powered Application Layer

Cursor, Loveable – for building tools. Start here only when you have a specific "what."

Perplexity

Use when you need sourced, current information – not model recall. Different tool, different job.

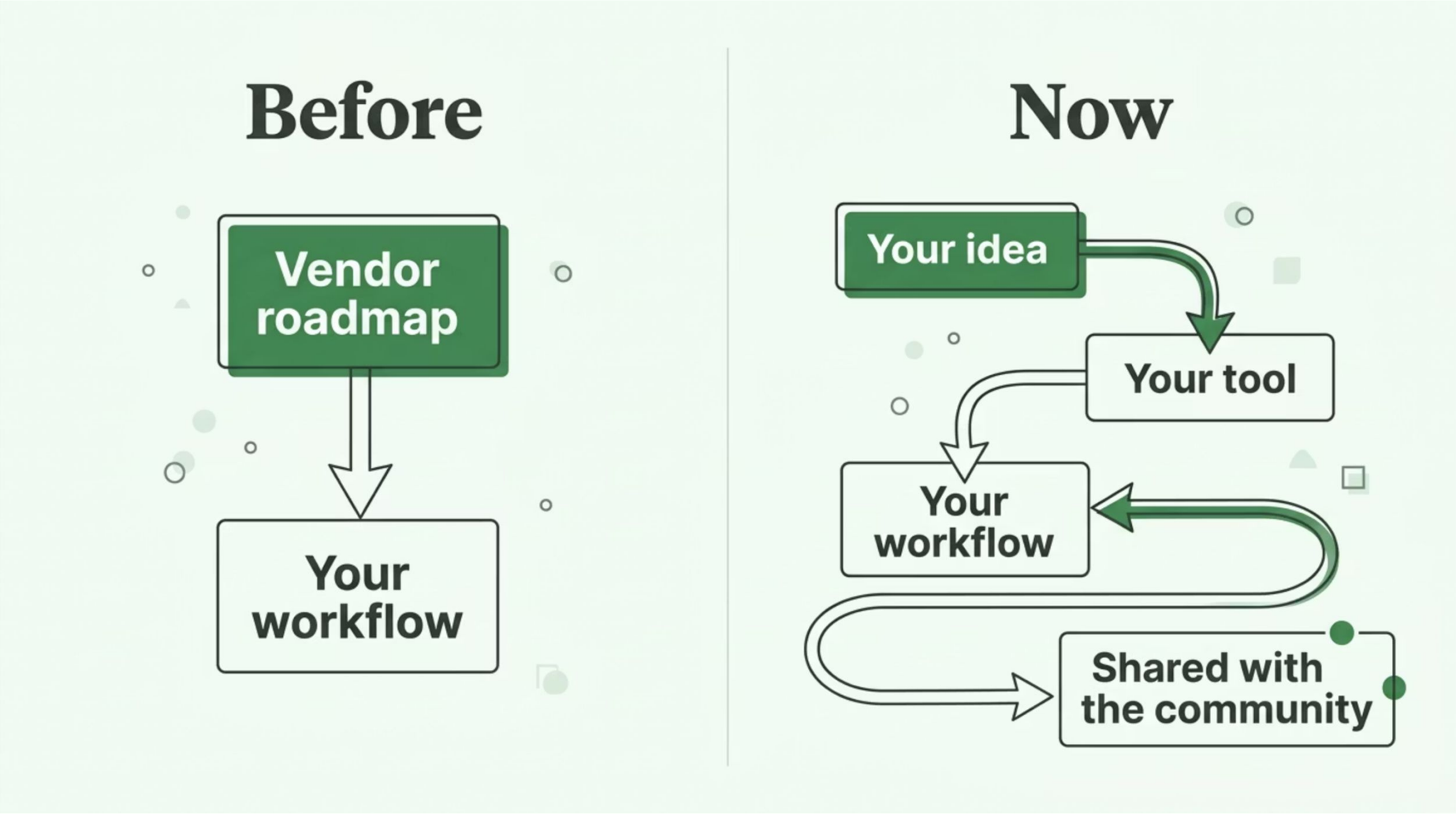
Steel Man Prompting

Ask GenAI to build the strongest version of the argument you disagree with. Forces rigor.

Ask Clarifying Questions First

Before a complex task, ask GenAI what it needs to know. Highest-leverage habit in the stack.

Stop Waiting for Vendors. Build in the Open.



- The era of waiting for big vendors to define your tools is over

Gneiss Ride and Block Party exist because someone **wanted** them, not because a PM approved them

- MVPs only – start embarrassingly small, one problem, one tool
- Share what you build – prompts, notebooks, architecture – even when you can't share the data
- **The more we build in the open, the faster we all level up. That compounds.**

You won't be replaced by AI.

You'll be replaced by a geoscientist who uses it.

The good news: that geoscientist is in this room.

Just Start

Open NotebookLM this week. Load some papers you've been meaning to read.

Describe one problem you've always wanted to solve. Ask GenAI how you'd approach it.

Let's Chat – I'll stay as long as anyone wants to keep the conversation going.

Thank You

Rocky Mountain Association of Geologists 2026

The Build Workflow: What To Do, When, and With What

PHASE 1 — SET UP YOUR BUILDING BUDDY

What you do: Write a persona prompt that defines your collaborator before any building starts. Match the persona to the task.

- Head Data Scientist: for analytical pipelines and notebooks
- Head Product Engineer: for simple apps and interfaces
- Solutions Architect: for complex multi-component systems

Why it matters: The model reasons differently depending on the role you assign. A poorly defined collaborator gives you a poorly defined build.

🔴 Model Horsepower: HEAVY – use your best model here. This prompt shapes everything downstream.

PHASE 2 — DESCRIBE

What you do: Go into precise detail about exactly what you want to build. This is your PRD moment – scope it ruthlessly.

- What is the input? What is the output?
- What does done look like?
- What is explicitly out of scope?

Lesson learned: Block Party v1 failed here. PRD was too ambitious – model built a mammoth full tool in the wrong direction. Backed up, went MVP-zero with Jupyter notebooks instead. Start smaller than you think you need to.

🔴 Model Horsepower: HEAVY – still your best model. Garbage in, garbage out.

PHASE 3 — ARCHITECT

What you do: Ask your heavy model to design the structure before anyone writes a line of code.

- What are the components?
- What builds first?
- Where are the dependencies?
- What could go wrong?

Why it matters: Architecture decisions made here are expensive to undo later. Spend the tokens now.

🔴 Model Horsepower: HEAVY – this is the highest-value use of your big model budget.

PHASE 4 — BUILD

What you do: Hand discrete, well-scoped features or MVP segments to smaller, faster, cheaper models to execute.

- One feature at a time
- One MVP at a time
- Never the whole thing at once

Gneiss Ride lesson: Went through 3-4 mini-PRD cycles, each scoping a single feature, each producing a cleaner result. Iteration beats ambition every time.

🟡🟢 Model Horsepower: LIGHT TO MEDIUM – smaller models handle execution well when the architecture is already defined. Save your heavy model budget for thinking, not typing.

Note: You can always escalate to a heavier model on a stubborn feature – it will improve the output but burn tokens faster. Know your budget.

PHASE 5 — FIX AND REPEAT

What you do: Things will break. This is not failure – this is the process.

- Describe the error back to the model in plain language
- Ask it to explain what went wrong before asking it to fix it
- If you've been going in circles for more than three cycles, back up and re-architect
- Systematic errors are findable. Random errors are the danger.

🟡 Model Horsepower: LIGHT – small fast models for debugging loops. Escalate only if the problem is architectural.

PHASE 6 — QC

What you do: No model. Just you.

- Test against data you already know the answer to
- If it's wrong in the same direction repeatedly, that's a systematic error – find it
- If it looks right but feels wrong, trust the feeling and investigate
- You caught the Jurassic anomaly. You caught the GPS drift. The model didn't.

🔵 Model Horsepower: NONE – this is the human layer. It is non-negotiable and it is not optional.

Key line: "The model will always give you an answer. Only you know if it's the right one."

MODEL HORSEPOWER LEGEND

🔴 Heavy model (Claude Opus, GPT-4o, Gemini Ultra) – thinking, persona, architecture, describe
🟡 Medium model – complex feature builds, persistent bugs
🟢 Light model (Claude Haiku, GPT-4o-mini) – execution, iteration, debugging loops
🔵 Human only – QC, validation, domain judgment. No exceptions.